A Watermark for Large Language Models

Kirchenbauer et al. - University of Maryland, ICML 2023
Reproduction and extensions

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Prof. Ali Kakhbod

MFE 230ZB Generative AI and Causality · October 2026

Two emails to Professor Kakhbod

To book a Zoom meeting about our fall project, I emailed Professor Kakhbod. Here are two versions.

A

Hi Professor Kakhbod,

Romain, Elias and I settled on the LLM watermarking paper for our fall project. Would you have 20 minutes on Zoom this week to go over our plan? We're flexible on Thursday and Friday.

Thanks,
Piero

B

Hi Professor Kakhbod,

We wanted to let you know that our project for fall is focused on LLM watermarking. We thought it would be great if we could discuss our plan next Tuesday morning in Zoom for 20 minutes. Option of Thursday or Friday works for us.

Thanks,
Piero

One of them I wrote with AI. **Which one?**

How would Professor Kakhbod catch me?

The naive approach

The naive approach: train a **classifier** on human vs AI text. Three problems:

- no guarantee and no known error rate;
- it degrades as models improve;
- it accuses humans.

	AI text caught	human text flagged as AI
OpenAI AI Text Classifier (January 2023)	26%	9%

OpenAI withdrew its classifier in July 2023 for its low rate of accuracy.

The paper's idea: a watermark

The paper

> A **watermark** is "a hidden pattern in text that is imperceptible to humans, while making the text algorithmically identifiable as synthetic."

The model owner adds it while the model writes. Anyone holding the key can check. Nothing is retrained.

Five properties

1. Detectable without the model or its API.
2. Works on any existing model, no retraining.
3. Detectable from any contiguous slice of text.
4. Cannot be removed without changing a large share of the tokens.
5. Comes with a rigorous p-value.

	AI text caught	human text flagged as AI
OpenAI AI Text Classifier (2023)	26%	9%
Watermark, at the paper's threshold $z = 4$	**98.4%**	**about 0.003%** (3 in 100,000)

Back to my emails, in colour

The paper

1. A secret key colours every token green or red, based on the token before it. Half the vocabulary is green.
2. A human does not know the key, so their tokens land on green only by luck: about 1 in 2.
3. The AI was pushed toward green while writing, so green piles up.

B: 50 green where chance gives 31, $z = 4.8$. By chance: 1 in 1.4 million.

A green tokens: 32 of 55, chance gives 28 ↩ = line break, a token too

Hi Professor K akh b od , ↩ R omain , Elias and I settled on the L LM watermark ing paper for our fall project . Would you have 2 0 minutes on Zoom this week to go over our plan ? We 're flexible on Thursday and Friday . ↩ Thanks , ↩ P iero

B green tokens: 50 of 62, chance gives 31

Hi Professor K akh b od , ↩ We wanted to let you know that our project for fall is focused on L LM watermark ing . We thought it would be great if we could discuss our plan next Tuesday morning in Zoom for 2 0 minutes . Option of Thursday or Friday works for us . ↩ Thanks , ↩ P iero

Since August 2026, Anthropic watermarks Claude's text with a scheme from this family (a version of SynthID-Text).

Algorithm 1: ban half the vocabulary

The paper · Algorithm 1 · generation

0 Tokenize — Text → token ids (integers from 0 to 50,264).

1 Run the model — A probability for each of the 50,265 tokens.
Unchanged: this is how any LLM works.

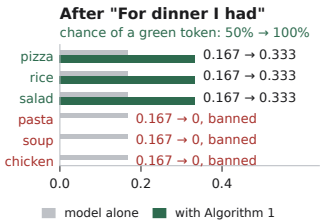
the watermark

2 Hash — hash(key, **previous token** id) → seed
First word: the last token of the prompt.

3 Random split — pseudo-random generator(seed) →
a random but **fixed** split of the whole vocabulary: **green** half, **red** half.

4 Ban and draw — **Red** → probability 0. Draw the next token from **green**.

5 Loop — The new token becomes the previous token. Back to 1.



Same key and same previous token: **same split**. Anyone with the key rebuilds every split **from the text alone**.

Live Demo!

Detection is a count against chance

The paper · Algorithm 1 · detection

0 Tokenize Same tokenizer: the same token ids.

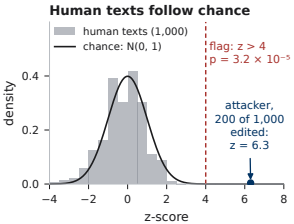
1 Rebuild For each token: hash(key, previous token id) → seed → the same split. Green or red? Anywhere in the text. No model, no prompt.

2 Count $|s|_G$ green tokens out of T.

3 Test H_0 H_0: the writer did not know the lists, so each token is green with probability $\frac{1}{2}$:

$$|s|_G \sim \text{Binomial}\left(T, \tfrac{1}{2}\right), \qquad z = \frac{|s|_G - T/2}{\sqrt{T}/2}$$

$z > 4$: **we reject H_0**, the text is watermarked. Under H_0 this happens with probability 3.2×10^{-5}: about 3 human texts in 100,000.



Our 1,000 human texts: none above 4.

All green: $z = \sqrt{T}$, so **16 tokens** are enough.

One edit turns at most **2 tokens** red: 200 edits on 1,000 tokens still give $z \approx 6.3$.

Where the hard rule breaks

The paper · Algorithm 1

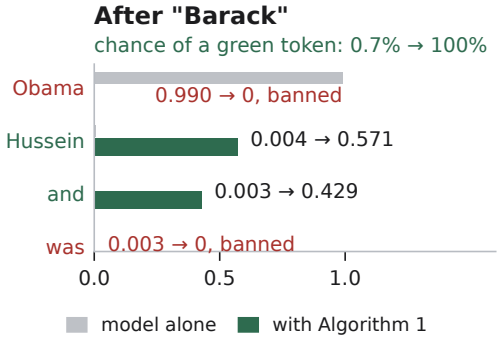
Low-entropy text: the next token is almost forced.

- After “Barack” the model puts about 99% on “Obama”. If “Obama” is red, the ban forces “Hussein” or “and”.
- Same for code, `for(i=0;i<n;i++)`, and for memorised quotes.

Two problems

1. Humans and models write the same thing there, so no test can tell them apart.
2. Forcing a different token wrecks the text.

So the authors replace the ban by a push.



Live Demo!

Algorithm 2: push green up instead of banning red

Idea: keep every token allowed, but raise the probability of green tokens. Two knobs: γ, the green share of the vocabulary, and δ, the size of the push.

Before. The model gives a score (logit) l_k to every token, normalised by the softmax:

$$p_k = \frac{e^{l_k}}{\sum_i e^{l_i}}$$

After. Add δ to every green logit, then normalise again:

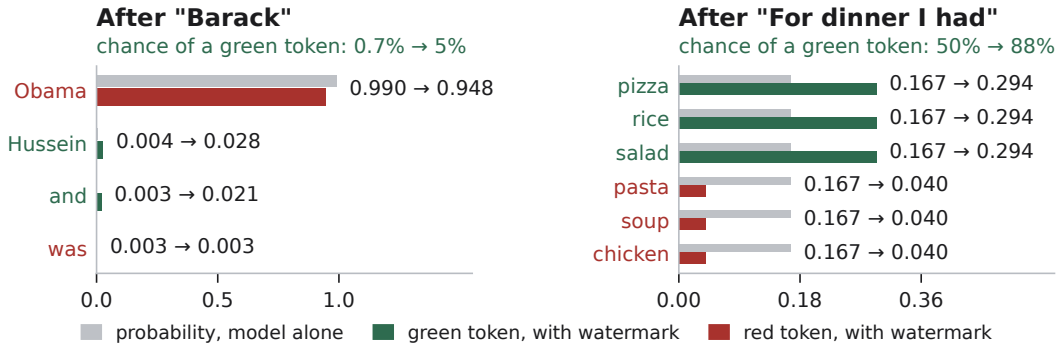
$$\hat{p}_k = \frac{e^{l_k + \delta\,\mathbf{1}\{k \in G\}}}{\sum_i e^{l_i + \delta\,\mathbf{1}\{i \in G\}}}$$

With $\alpha = e^{\delta}$ and $p_G = \sum_{i \in G} p_i$, the model's own mass on green:

$$\hat{p}_k = \frac{\alpha\, p_k}{1 + (\alpha - 1)\, p_G} \;\; (k \text{ green}), \qquad \hat{p}_j = \frac{p_j}{1 + (\alpha - 1)\, p_G} \;\; (j \text{ red})$$

The push only acts where the model has a choice

The paper · Algorithm 2



Confident model: the push changes almost nothing, so quality is kept. Several good options: the push decides, so green tokens pile up.

Detection: a test against chance

The paper · Algorithm 2

H_0: the text was written by a human, who does not know the key.

Each token is then green with probability γ, independently of the others.

LLN $$|s|_G \sim \mathrm{Binomial}(T, \gamma) \implies \frac{|s|_G}{T} \xrightarrow[T\to\infty]{\text{a.s.}} \gamma$$

CLT $$Z_T = \frac{|s|_G - \gamma T}{\sqrt{T\,\gamma(1-\gamma)}} \xrightarrow[T\to\infty]{d} \mathcal{N}(0,1)$$

Watermarked text: $|s|_G/T \to q > \gamma$, so $Z_T \approx \sqrt{T}\,\dfrac{q-\gamma}{\sqrt{\gamma(1-\gamma)}} \to \infty$.

How fast Z_T becomes Gaussian depends on γ (Berry–Esseen error $\propto \frac{\gamma^2+(1-\gamma)^2}{\sqrt{T\gamma(1-\gamma)}}$, 4 times larger at $\gamma = 0.05$ than at $\gamma = \frac{1}{2}$). Exact false-alarm rate at $Z_T > 4$, $T = 200$: 2.5×10^{-5} at $\gamma = \frac{1}{2}$, but 1.9×10^{-4} at $\gamma = 0.05$, six times the Gaussian value.

Live Demo!

Limit 1: when the model has no choice

The paper · Algorithm 2

The test works if $q > \gamma$. **What sets q?** How much the model hesitates.

Spike entropy of the model's next-token distribution:

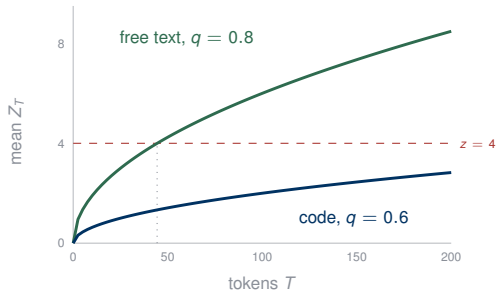
$$S(p) = \sum_k \frac{p_k}{1 + \zeta^\star p_k}$$

all mass on one token (code): $S = \frac{1}{1+\zeta^\star} = 0.57$

uniform over N tokens: $S = \frac{N}{N+\zeta^\star} \approx 1$

Theorem 4.2, with $S^\star$ the average of S along the text:

$$\mathbb{E}\,|s|_G \;\geq\; T \cdot c \cdot S^\star, \qquad c = \frac{\gamma\alpha}{1 + (\alpha - 1)\gamma} = 0.88$$



Tokens needed for $Z_T = 4$: **45** vs **400**.

Illustrative values, $\gamma = 0.5$, formula of the previous slide.

$\gamma = 0.5$, $\delta = 2$: $\zeta^\star = 0.76$, $c = 0.88$. Paper: $S^\star = 0.807$, so at least 142.2 green tokens of 200; observed 159.5. Real data in Extension 1.

Limit 2: other weak points, and how to remove the mark

Weak points of the test

- **Repeated pairs**: same pair, same colour. Under H_0, $|s|_G$ is no longer Binomial: a repetitive human text can be flagged.
- **Quality**: a larger δ is easier to detect but distorts the text more.
- **The key**: public, anyone can rebuild the lists; private, detection sits behind an API that must be rate-limited.

Removing the mark (paper, Sec. 7 and App. B)

attack	how	defence
Paraphrase	rewrite with another model (T5)	costs quality
Insert or delete	add or drop tokens, shifting the next lists	costs quality
Character tricks	homoglyphs, invisible characters	normalise before hashing
Emoji attack	ask for an emoji after each word, then delete them	train the model to refuse

Removing the mark is trivial if quality does not matter. The real question is its cost.

What the soft watermark leaves open

The paper

Limit	Why	What we test
Constrained text (code, short factual answers)	little choice, little signal	Extension 1
A modern rewrite	the paper attacks with T5 word swaps (2020)	Extension 2
A watermarked passage pasted into human text	the paper's conclusion lists it as open	Extension 3
Side effects of the key itself	the key fixes the colour of every token pair, "stop" included	Extension 4

Repeated phrases get the same colour every time, which breaks the independence the test assumes (backup B2, B14).

Same experiments, our own code

Reproduction

	Paper	Ours
Texts	C4 news articles: start = prompt, last 200 tokens = human text	same
Writing model	OPT-1.3B	same
Settings	$\tau = 0.7$, $T = 200 \pm 5$, γ and δ grids, multinomial and beam search	same
Quality judge	OPT-2.7B perplexity	same
Sample	about 500 texts per setting	100 to 500 per setting (497 for the headline), plus 1,000 human texts
Attack	T5-Large span replacement (code never released)	rebuilt from the paper's description
Hardware	not stated	one laptop, about 20 hours

First check: our green lists are identical to the official code on 1,000 random contexts (and 200 for Qwen). Any gap with the paper comes from the experiment, not from the code.

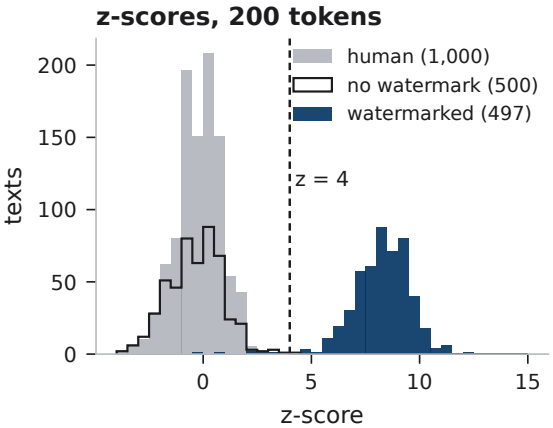
The headline reproduces

Reproduction

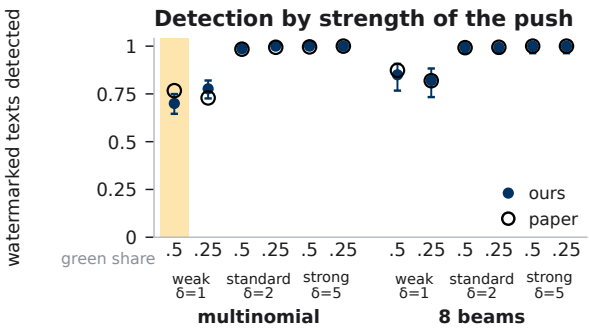
$\gamma = 0.5$, $\delta = 2$, $T = 200$	Paper	Ours
detected at $z = 4$	98.4%	98.6% [97.1, 99.3]
human texts flagged	0	0 of 1,000
mean green tokens	159.5	158.3
Theorem 4.2 bound	142.2	143.2
4-beam search detected	99.6%	100%

497 watermarked texts (150 with 4 beams) and 1,000 human texts.

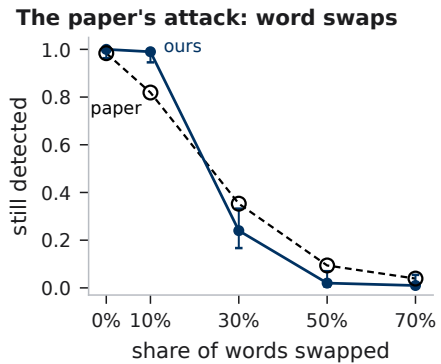
Brackets: 95% interval.



Error rates and the paper's attack reproduce



11 of 12 settings within our 95% intervals; the shaded one is the exception.



Same shape. A light attack removes less in our runs, heavier ones slightly more.

What we could not reproduce

Reproduction

1. **One weak-push setting detects less**: multinomial, $\delta = 1$, $\gamma = 0.5$, detected at $z = 5$: 0.37 [0.31, 0.42] vs 0.50. With a weak push, detection rests on a few frequent token pairs whose colour is fixed by the key; we did not test another key.
2. **The lightest T5 attack removes less**: at 10% of words swapped, 0.99 vs 0.82 still detected. The attack code was never released; we rebuilt it from the description. From 30% on, the results agree or come close.
3. **Perplexities slightly higher**: about +0.1 to +0.3 under multinomial sampling (one setting +1.7); same ordering everywhere.
4. **Short answers**: OPT-1.3B cannot do the task, so we used Qwen2.5-1.5B-Instruct instead of the paper's much larger models. No measurable accuracy drop from the watermark (exact match 0.202 to 0.212), where the paper reports about -0.04.

Free vs constrained text

Extension 1 · Entropy decides

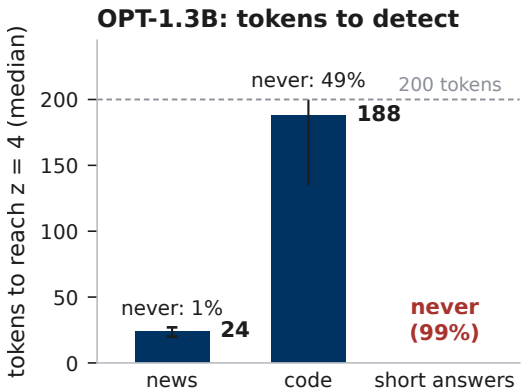
The paper admits that constrained text (code, factual answers: the next token is almost forced) is hard, but tests almost only on news.

We challenge the idea that news results describe real use.

Why: the push only acts where the model has a choice. Less freedom, less signal: Theorem 4.2's own prediction.

Result: median tokens to reach $z = 4$: news 24, code 188 (49% never in 200 tokens), short answers never.

OPT-1.3B. News (200 texts) and code (164): $\gamma = 0.25$, $\delta = 2$, sampling. Short answers (500): $\gamma = 0.5$, $\delta = 2$, greedy, as in the paper's Table 10.



The rewrite attack

Extension 2 · A 2026 attacker

The paper says removing the watermark takes many word swaps, which damage the text.

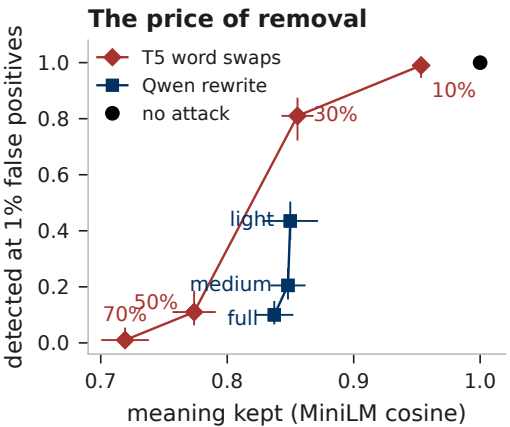
We challenge the attacker: today one prompt to a free AI model rewrites the text.

Why: the watermark lives in pairs of neighbouring tokens. A swap breaks few pairs; a rewrite breaks most.

Result, at the same meaning kept (0.85):

	T5, 30%	Qwen, light
detected, 1% false positives	81%	43.5%
AUC	0.995	0.881
model calls	60 to 140	1

AUC: 1 = always told apart from human text, 0.5 = coin flip.



A watermarked passage inside a human text

Extension 3 · Copy-paste

The paper tests texts watermarked from start to end; its conclusion lists the mixed case as open.

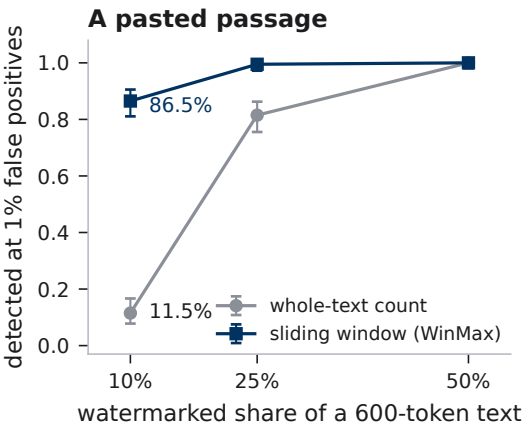
We challenge counting over the whole text when only a passage is pasted in.

Why: the human tokens around the passage pull the count back to chance. A sliding window restores the contrast.

Result, 60-token passage in a 600-token human text, 1% false alarms:

detector	detected
whole-text count	11.5%
sliding window (WinMax)	86.5%

WinMax: Kirchenbauer et al., ICLR 2024 (the authors' follow-up).



What we take away

Takeaways

- **The paper reproduces** with our own code: 98.6% vs 98.4% detected, 11 of 12 error-rate settings, green lists identical to the official code.
- **Three limits**: it fades on constrained text (code, short answers); a one-prompt rewrite by a free model removes it more efficiently than the paper's attack, though still at some cost in meaning; a pasted passage is lost in a whole-text count.
- **One fix that exists**: a sliding window recovers the pasted passage (86.5% vs 11.5%).
- **One surprise**: the key itself can change when the model stops.

> Anthropic notes its watermark is less reliable on short, factual or code text. Extension 1 shows why.

Next step (Elias): use the detector as a filter for sentiment data, and measure how much of a manipulated signal it restores.

Backup contents

Backup

The paper

B1 Spike entropy and Theorem 4.2 in full
B2 What the test assumes
B3 Strength vs quality (Fig. 2)
B4 Detection vs length (Fig. 3)

Reproduction

B5 Table 8 in full
B6 AUC and perplexity (Fig. 4, 8, 9)
B7 The theory curve (Fig. 7) and Extension 1a
B8 The T5 attack in full (Table 9)
B9 Short factual answers (Table 10)
B10 What a hit and a miss look like
B11 False alarms without a watermark
B12 Where our numbers differ, and our deviations

Extensions

B13 Extension 1, every cell
B14 Repetition breaks the test on code
B15 The theorem, text by text
B16 Models tuned to follow instructions (1c)
B17 Extension 2, every attack
B18 Extension 2 at the paper's threshold
B19 Extension 3 in full
B20 Extension 4 in full

Method and code

B21 Code, compute and reproducibility

Click a line to jump; the word "Backup" under each title returns here.

Spike entropy and Theorem 4.2 in full

Backup

Spike entropy of p with modulus ζ:

$$S(p, \zeta) = \sum_k \frac{p_k}{1 + \zeta p_k}$$

from $1/(1+\zeta)$ (one token has all the mass) to close to 1 (mass spread).

$$\zeta^\star = \frac{(1-\gamma)(\alpha-1)}{1+(\alpha-1)\gamma}, \qquad c = \frac{\gamma\alpha}{1+(\alpha-1)\gamma}$$

Theorem 4.2, $S^\star$ = average spike entropy at $\zeta^\star$:

$$\mathbb{E}\,|s|_G \geq TcS^\star, \qquad \mathrm{Var}\,|s|_G \leq TcS^\star(1 - cS^\star)$$

The paper writes the modulus as z; we write ζ to avoid confusion with the z-score.

At $\gamma = 0.5$, $\delta = 2$: $\zeta^\star = 0.76$, $c = 0.881$.

$T = 200$	Paper	Ours
average spread $S^\star$	0.807	0.813
bound on $\mathbb{E}\,\lvert s\rvert_G$	142.2	143.2
observed mean	159.5	158.3
bound on σ	6.41	6.38
observed σ	–	9.85
predicted detection	98.6%	99.0%

The σ bound holds at a fixed entropy level; across texts the entropy varies, so the observed spread is wider.

What the test assumes

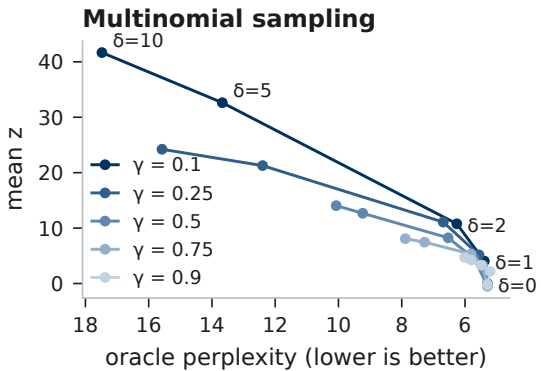
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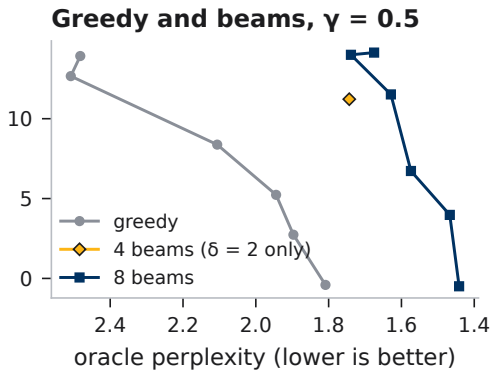
1. **Independent tokens.** Repeated phrases get the same colour every time, so they count many times (B11, B14).
2. **Random lists.** The lists are pseudo-random and fixed by the key. Across six keys, the mean z of unwatermarked model text ranges from -0.52 to $+0.71$ (human text: -0.36 to $+0.55$), with standard deviations 1.03 to 1.36. With our key, false alarms at $z = 4$ stay at 0 of 1,000 human texts.
3. **Long enough text** for the normal approximation. Short answers (16 tokens or fewer) give z only a few possible values.

> Decoding and re-encoding the text moves z by 0.005 on average. Ignoring repeated pairs, headline detection is 97.2% [95.3, 98.3] instead of 98.6%.

Strength vs quality (Fig. 2)

Backup

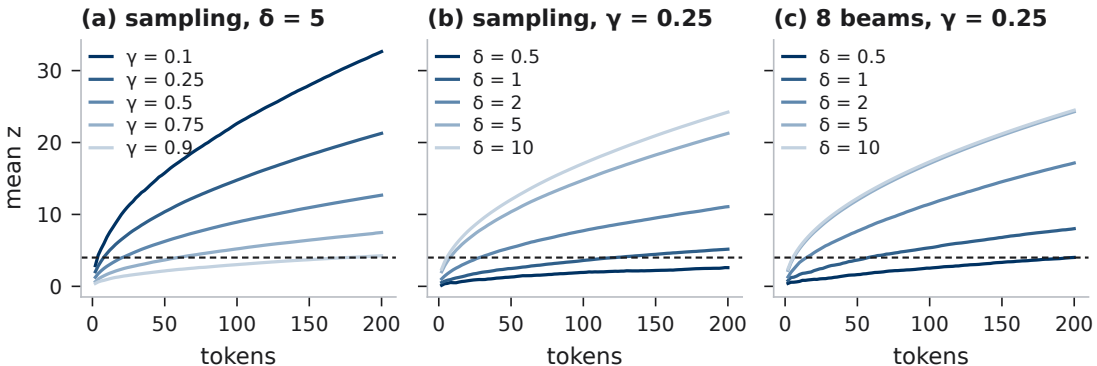




Smaller γ is better at the same perplexity ($\delta = 2$: $z = 10.8$ at $\gamma = 0.1$ vs 8.3 at $\gamma = 0.5$, perplexity 6.3 vs 6.5). With 8 beams, $z = 11.5$ at perplexity 1.63 (1.44 without watermark).

Detection vs length (Fig. 3)

Backup



First length where the mean z exceeds 4: (a) 4 / 8 / 22 / 60 / 176 tokens for γ = 0.1 / 0.25 / 0.5 / 0.75 / 0.9; (b) 124 / 30 / 8 / 6 for δ = 1 / 2 / 5 / 10; (c) 16 at $\delta = 2$, and mean $z > 5$ from 24 tokens (paper: about 35).

Table 8 in full

Backup

sampling	δ	γ	n	TPR $z = 4$ [95% CI] (paper)	verdict	TPR $z = 5$ [95% CI] (paper)	verdict
multinomial	1	0.50	300	0.700 [0.646, 0.749] (0.767)	borderline	0.367 [0.314, 0.423] (0.504)	**different**
multinomial	1	0.25	300	0.777 [0.726, 0.820] (0.729)	consistent	0.537 [0.480, 0.592] (0.482)	consistent
multinomial	2	0.50	497	0.986 [0.971, 0.993] (0.984)	consistent	0.980 [0.963, 0.989] (0.978)	consistent
multinomial	2	0.25	300	1.000 [0.987, 1.000] (0.994)	consistent	1.000 [0.987, 1.000] (0.988)	consistent
multinomial	5	0.50	300	1.000 [0.987, 1.000] (0.996)	consistent	1.000 [0.987, 1.000] (0.992)	consistent
multinomial	5	0.25	300	1.000 [0.987, 1.000] (1.000)	consistent	1.000 [0.987, 1.000] (0.998)	consistent
8 beams	1	0.50	100	0.850 [0.767, 0.907] (0.873)	consistent	0.750 [0.657, 0.825] (0.812)	consistent
8 beams	1	0.25	100	0.820 [0.733, 0.883] (0.819)	consistent	0.750 [0.657, 0.825] (0.770)	consistent
8 beams	2	0.50	150	0.993 [0.963, 0.999] (0.992)	consistent	0.987 [0.953, 0.996] (0.984)	consistent
8 beams	2	0.25	150	0.993 [0.963, 0.999] (0.994)	consistent	0.987 [0.953, 0.996] (0.990)	consistent
8 beams	5	0.50	100	1.000 [0.963, 1.000] (1.000)	consistent	1.000 [0.963, 1.000] (1.000)	consistent
8 beams	5	0.25	100	1.000 [0.963, 1.000] (1.000)	consistent	1.000 [0.963, 1.000] (1.000)	consistent

False positive rate at $z = 4$: 0 [0, 0.004] on 1,000 human texts in every row. On 500 unwatermarked model texts: 0.002 (one text, repetitive; B11).

Verdict: consistent = paper inside our interval; borderline = outside, but the paper's own interval overlaps ours; different = no overlap.

AUC and perplexity (Fig. 4, 8, 9)

Backup

Multinomial sampling

δ	γ	AUC (paper)	perplexity (paper)
1	0.50	0.997 (0.985)	5.72 (5.4)
1	0.25	0.992 (0.989)	5.56 (5.5)
2	0.50	0.999 (0.998)	6.53 (6.2)
2	0.25	1.000 (0.998)	6.69 (6.6)
5	0.50	1.000 (1.000)	9.23 (9.1)
5	0.25	1.000 (1.000)	12.40 (10.7)
no watermark			5.29 (5.1)

8 beams

δ	γ	AUC (paper)	perplexity (paper)
1	0.50	0.991 (0.987)	1.57 (1.2)
1	0.25	0.986 (0.977)	1.54 (1.2)
2	0.50	1.000 (0.999)	1.63 (1.2)
2	0.25	0.998 (1.000)	1.70 (1.3)
5	0.50	1.000 (1.000)	1.74 (1.2)
5	0.25	1.000 (1.000)	1.68 (1.3)
no watermark			1.44 (1.2)

Human text, same judge: perplexity 11.06. Our perplexities sit slightly above the paper's (+0.1 to +0.3 with sampling, +0.3 to +0.5 with 8 beams), except $\delta = 5$, $\gamma = 0.25$ (+1.7); same ordering everywhere.

The theory curve (Fig. 7) and Extension 1a

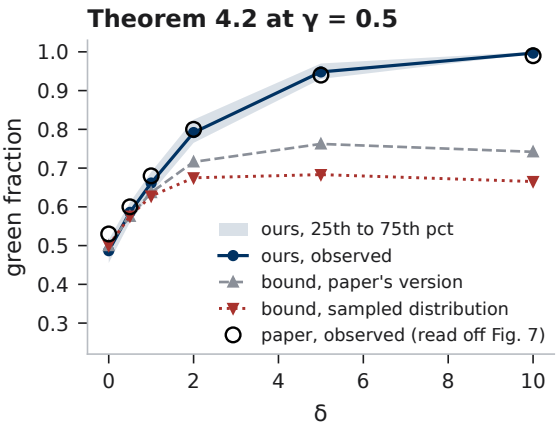
Backup

The code adds δ before dividing by the temperature 0.7, so the distribution actually sampled sees δ/τ.

We recompute the bound on that distribution (spike entropy of $\mathrm{softmax}(I/\tau)$, $\alpha = e^{\delta/\tau}$).

It holds in all 22 runs, but is looser for $\delta \geq 2$ ($\gamma = 0.5$, $\delta = 2$: 0.675 vs 0.716; observed 0.792).

So the paper's large-δ gap is the bound's own conservatism, not the temperature.



The T5 attack in full (Table 9)

Backup

ε	TPR $z = 4$ [95% CI] (paper)	verdict	TPR $z = 5$ [95% CI] (paper)	verdict
0	1.000 [0.963, 1.000] (0.984)	consistent	0.980 [0.930, 0.994] (0.977)	consistent
0.1	0.990 [0.946, 0.998] (0.819)	**different**	0.870 [0.790, 0.922] (0.577)	**different**
0.3	0.240 [0.167, 0.332] (0.353)	borderline	0.090 [0.048, 0.162] (0.127)	consistent
0.5	0.020 [0.006, 0.070] (0.094)	**different**	0.000 [0, 0.037] (0.029)	consistent
0.7	0.010 [0.002, 0.054] (0.039)	consistent	0.000 [0, 0.037] (0.012)	consistent

ε	AUC (paper)	perplexity (paper)	tokens after	words replaced
0	1.000 (0.998)	6.7 (6.3)	200	0
0.1	1.000 (0.988)	9.2 (13.1)	201	20.0
0.3	0.996 (0.954)	16.0 (21.2)	202	60.0
0.5	0.868 (0.838)	24.8 (28.4)	206	99.9
0.7	0.652 (0.696)	33.5 (33.9)	210	130.4

Single-word variant (each replacement is exactly one word, as the paper describes): multinomial, $\gamma = 0.5$, $\delta = 2$, 100 texts. AUC against the 1,000 unattacked human texts.

Multi-word variant (T5 may fill a span with several words), 200 texts: a weaker attack, because texts grow (295 tokens at 30%) and z grows with $\sqrt{T}$: 0.495 vs 0.24 detected at 30%.

Attack code never released. Not run: T5 on beam-search texts.

Short factual answers (Table 10)

Backup

	exact match		mean z	
model	no watermark	watermark	no watermark	watermark
FLAN-UL2 (paper)	0.374	0.336	-0.007	0.402
BLOOMZ (paper)	0.296	0.259	0.008	0.255
OPT-1.3B	0.000	0.004	-0.414	1.432
Qwen2.5-1.5B	0.072	0.070	-0.236	1.183
Qwen2.5-1.5B-Instruct	0.202	0.212	-0.053	0.803

TriviaQA validation, first 500 questions, greedy, up to 16 new tokens, $\gamma = 0.5$, $\delta = 2$, the paper's prompt. EM = exact match against the normalised answer aliases.

Qwen2.5-1.5B-Instruct: 0.202 [0.169, 0.239] $\rightarrow$ 0.212 [0.178, 0.250], no measurable change; answers average 5.8 tokens. OPT-1.3B cannot do the task with this prompt.

What a hit and a miss look like

Backup

Hit (spread 0.875, $z = 9.62$). Prompt: … While similar modern devices do work successfully (as seen in the NOVA

Watermarked: series of weather forecasts, or in the GPS unit of the iPod) the disks used in the sagas were not equipped with a bearing system or with a complete circle, much less a complete star-track. …

Miss (spread 0.571, $z = 0.57$). Prompt: … He said the construction phase would create

Watermarked: jobs in the region. “We’re very supportive of the announcement, it’s a fantastic facility to have in Tallangatta,” Cr Scales said. Once open, 24 existing DELWP staff, …

The human continuation is word for word the same: the model copies the article, so there is no choice to push.

False alarms without a watermark are repetitive text

Backup

The only unwatermarked ($\delta = 0$) model texts that cross $z = 4$, one per γ at most, out of 500:

γ	z	z, repeats ignored	text
0.10	4.01	3.13	. . . your average number of rejections is between one and five . . .
0.25	4.08	2.94	. . . It's a great role for Dern, and he does an excellent job . . .
0.50	4.38	0.40	. . . Bill Evans, Herbie Hancock, Miles Davis, Stan Getz, Art Garfunkel, John Coltrane, Bill Evans, Herbie Hancock, Herbie Hancock, . . .

Human texts: 0 of 1,000 cross $z = 4$ at every γ (mean z -0.25 to $+0.09$, standard deviation 1.04 to 1.10). Unwatermarked model texts are more spread out (standard deviation 1.27 to 1.35) because models repeat themselves.

Where our numbers differ, and our deviations

Backup

Differences, with intervals

- Table 8, multinomial $\delta = 1$, $\gamma = 0.5$: borderline at $z = 4$ (0.70 vs 0.77), different at $z = 5$ (0.37 [0.31, 0.42] vs 0.50).
- T5, single word: different at 10% (both thresholds) and at 50% ($z = 4$: 0.02 vs 0.094); borderline at 30% ($z = 4$); consistent elsewhere.
- Perplexity: largest gap $\delta = 5$, $\gamma = 0.25$, 12.4 vs 10.7.
- Short answers: no measurable EM drop (paper about -0.04), with a different model.
- Headline mean green count 158.3 [157.5, 159.2] vs 159.5, just outside.

Deviations from the paper

- Green lists from a CPU random generator (the paper: CUDA).
- Prompts longer than 2,048 tokens: kept their last 1,847, not the first.
- Length filter per setting, not on watermarked and unwatermarked pairs; the headline kept 497 of 1,000.
- Greedy news decoding suppresses end-of-text, like beam search.
- T5 in bfloat16, words = `\w+` runs, $\pm$60-token windows, nested ε levels.
- Qwen news prompts: last 150 words of the article, for both models.
- Paraphraser: temperature 0.7, top-k 0, top-p 1 (not Qwen's chat defaults).

Extension 1, every cell

Backup

			entropy,	TPR at 1% FPR		tokens to reach $z = 4$	
domain	model	n	no watermark	repeats counted	repeats ignored	median	never
news	OPT	200	2.41	0.990	0.990	24 [20, 27]	0.010
code	OPT	164	0.66	0.524	0.616	188 [135, never]	0.494
short answers	OPT	500	2.06	0.518	0.698	never	0.990
news	Qwen	198	2.52	0.995	0.995	25 [24, 27]	0.015
code	Qwen	164	0.59	0.244	0.512	never	0.598
short answers	Qwen	500	1.74	0.142	0.156	never	0.998
news	Qwen-Instruct	200	1.96	0.990	0.990	27 [24, 29]	0.040
code	Qwen-Instruct	164	0.43	0.457	0.488	never	0.732
short answers	Qwen-Instruct	500	1.18	0.054	0.052	never	0.998
news, $\delta = 4$	Qwen-Instruct	200	1.96	0.990	0.990	8 [8, 8]	0.010

OPT = OPT-1.3B, Qwen = Qwen2.5-1.5B, Qwen-Instruct = Qwen2.5-1.5B-Instruct. News and code: $\gamma = 0.25$, $\delta = 2$, sampling at $\tau = 0.7$; short answers: $\gamma = 0.5$, $\delta = 2$, greedy. Entropy: mean Shannon entropy (nats) of the unwatermarked model's next-token distribution. The 1% FPR threshold comes from each cell's own unwatermarked texts.

Repetition breaks the test on code

Backup

Code repeats the same token pairs (newline then indentation, dozens of times). Each repeat gets the same colour, so one pair can dominate the count.

code, $\gamma = 0.25$, $\delta = 2$	repeats counted	repeats ignored
OPT-1.3B, mean z without watermark	-2.88	-0.45
OPT-1.3B, TPR at 1% FPR	0.52 [0.45, 0.60]	0.62 [0.54, 0.69]
Qwen2.5-1.5B, TPR at 1% FPR	0.24 [0.19, 0.32]	0.51 [0.44, 0.59]

> The paper's own remedy (its Appendix B): count each repeated pair once. It restores calibration and doubles Qwen's detection.

The theorem, text by text

Backup

		green fraction			texts below the prediction	
domain	model	observed	predicted, paper-style	predicted, corrected	paper-style	corrected
news	OPT	0.584	0.494	0.453	4.0%	2.5%
code	OPT	0.372	0.326	0.300	40.8%	32.9%
short answers	OPT	0.679	0.679	0.679	57.6%	57.6%
news	Qwen	0.597	0.506	0.466	3.0%	0.5%
code	Qwen	0.382	0.342	0.320	27.4%	20.7%
short answers	Qwen	0.787	0.701	0.701	24.8%	24.8%
news	Qwen-Instruct	0.572	0.493	0.458	4.0%	1.5%
code	Qwen-Instruct	0.341	0.313	0.299	27.4%	18.9%
short answers	Qwen-Instruct	0.693	0.648	0.648	37.6%	37.6%
news, $\delta = 4$	Qwen-Instruct	0.847	0.658	0.542	0.5%	0.5%

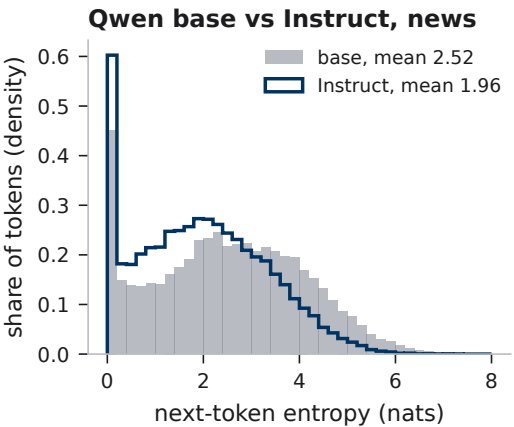
Prediction: Theorem 4.2 from each text's own mean spike entropy, as a green fraction. On average it sits below the observed value in every cell (news, OPT-1.3B: 0.584 vs 0.453 corrected), except short answers with OPT-1.3B, where the two are equal. Single texts often fall below it on code and short answers.

Models tuned to follow instructions (Extension 1c)

Backup

Qwen2.5-1.5B vs its Instruct version: same architecture and tokenizer.

- Instruct is more confident: entropy 2.52 $\rightarrow$ 1.96 nats on news, 0.59 $\rightarrow$ 0.43 on code, 1.74 $\rightarrow$ 1.18 on short answers.
- Yet news detection is the same: 99.5% vs 99.0% at 1% false positives. The watermark pushes Instruct off its confident path (entropy 1.96 $\rightarrow$ 2.30 under the watermark).
- It pays in quality: perplexity 5.7 $\rightarrow$ 9.5 (OPT-1.3B: 5.2 $\rightarrow$ 6.9).
- $\delta = 4$ cuts the tokens needed from 27 to 8, at perplexity 25.6.
- On code, no consistent base vs Instruct difference.



Extension 2, every attack

Backup

attack	meaning kept	tokens changed	green fraction	TPR $z = 4$	TPR 1% FPR	AUC	perplexity
none (re-encoded)	1.000	0.00	0.79	1.000	1.000	1.000	6.6
T5 single word, 10%	0.953	0.11	0.72	0.990	0.990	1.000	9.2
T5 single word, 30%	0.856	0.34	0.62	0.240	0.810	0.995	16.0
T5 single word, 50%	0.774	0.57	0.55	0.020	0.110	0.824	24.8
T5 single word, 70%	0.719	0.75	0.51	0.010	0.010	0.607	33.5
T5 multi-word, 10%	0.951	0.22	0.72	0.955	1.000	1.000	8.1
T5 multi-word, 30%	0.881	0.65	0.62	0.495	0.925	0.991	9.9
Qwen rewrite, light	0.850	0.65	0.60	0.220	0.435	0.881	30.1
Qwen rewrite, medium	0.848	0.81	0.55	0.035	0.205	0.773	21.1
Qwen rewrite, full	0.837	0.86	0.53	0.015	0.100	0.705	20.6

An instruction to “lightly edit” already rewrites 65% of the tokens. Rewrites score high perplexity because they lose the link to the prompt (texts often restart a sentence).

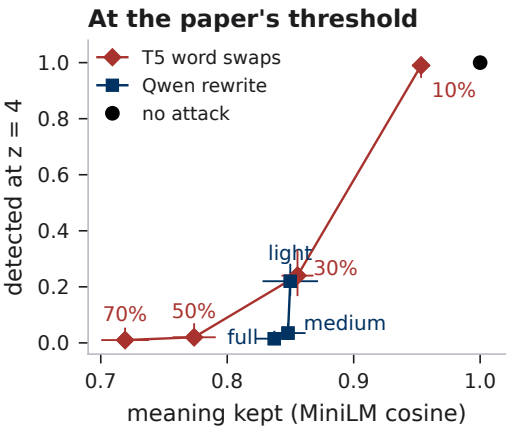
Extension 2 at the paper's fixed threshold

Backup

Same attacks, detection at the paper's fixed threshold $z = 4$ instead of a calibrated 1% false-positive rate.

At matched meaning the light rewrite and the word swaps are equal: 0.22 [0.17, 0.28] vs 0.24 [0.17, 0.33].

The gap only appears at a calibrated threshold: word-swapped texts keep many z-scores between the calibrated threshold (about 2.4) and 4; rewritten ones do not.



Extension 3 in full

Backup

600-token documents of human C4 text with one watermarked passage ($\gamma = 0.5$, $\delta = 2$) at a random position; 200 documents per share.

watermarked share	whole-text count	sliding window (WinMax)	mean z	mean WinMax
10% (60 tokens)	0.115 [0.078, 0.167]	0.865 [0.811, 0.906]	1.03	4.98
25% (150 tokens)	0.815 [0.755, 0.863]	0.995 [0.972, 0.999]	3.30	7.36
50% (300 tokens)	1.000 [0.981, 1.000]	1.000 [0.981, 1.000]	6.86	10.25

Detection at a 1% false-alarm rate. Thresholds from 1,000 human-only documents: whole text $z > 2.20$, WinMax > 4.02 (human means -0.37 and 3.03). At the paper's $z = 4$ the whole-text count catches only 1%, 28.5% and 97.5%.

Extension 4 in full

Backup

hash key	EOS green after "."	P(EOS) multiplier after "."	reach 195 tokens	stop before 50 tokens
15485863 (ours)	yes	2.29	0.53 [0.43, 0.63]	13% [8, 21]
15485867	no	0.17	0.80 [0.71, 0.87]	3% [1, 8]
32452843	yes	1.18	0.76 [0.67, 0.83]	1% [0, 5]
49979687	no	0.32	0.87 [0.79, 0.92]	1% [0, 5]
no watermark	–	1	0.75 [0.66, 0.83]	4% [2, 10]

$\gamma = 0.5$, $\delta = 2$, the same first 100 prompts per key. Full sample, our key: 16% [14, 18] stop before 50 tokens vs 6% [4, 7] without watermark (1,000 and 711 texts). At $\delta \geq 5$ and γ 0.5 to 0.75: 29% to 37%. At $\gamma = 0.25$ texts stop *less* often than without watermark (2.8% at $\delta = 2$, 2.3% at $\delta = 5$): "stop" is red after "." there.

Code, compute and reproducibility

Backup

Repository: `github.com/pieropls/230ZB`

Pipeline: prompts → generation with the watermark processor → detection → attacks → statistics → figures → notebook.

The notebook regenerates every table and figure from saved results. A full re-run takes about 20 hours on one laptop.

Apple M5 Pro, 24 GB, PyTorch on `mps`, fp16 (T5 in bfloat16), transformers 5.18.

step	seconds
OPT-1.3B, sampling, per text	0.59 to 0.84
OPT-1.3B, 4 beams, per text	7.1
OPT-1.3B, 8 beams, per text	9.2
oracle perplexity, per text	0.31
Qwen generation, per text	0.52 to 0.53
Qwen rewrite, per text	0.63
T5 attack, one round of 32 texts	9.5